Practice workbook · Explain an enzyme practical result, including levelling off as substrate runs out

Observations such as falling pH, a positive sugar test, a clearing film or a plateau in product, explained by digestion and by substrate becoming limiting.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ENZYME-RESULT-EXPLAIN

1. Question 7 · part b.ii · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

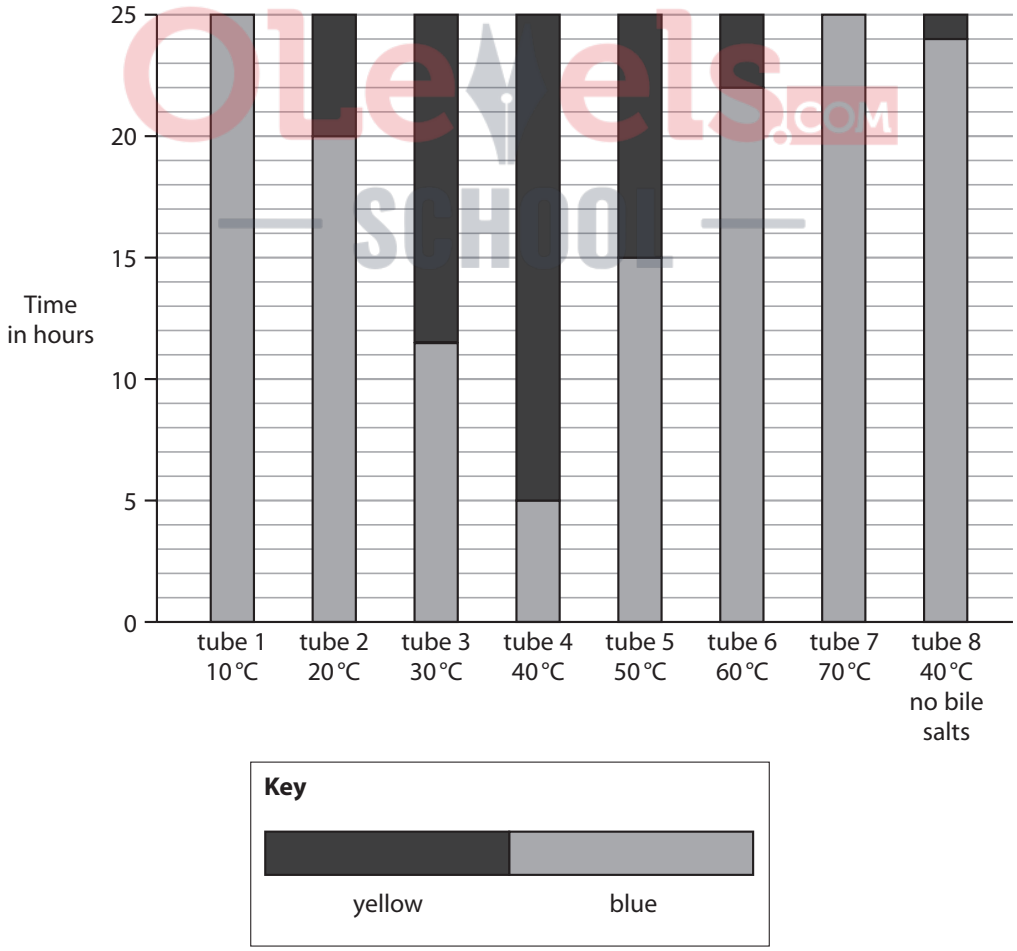
7 A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.



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1. Question 7 · part b.ii · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

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(a) (i) State three factors that should be kept constant in the student's investigation.

(3)

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

(1)

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(iii) Discuss why this temperature is only an approximate optimum temperature.

(3)

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(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

(2)

time = ... hours

(ii) Explain why the pH decreases during the 25 hours.

(3)

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1. Question 7 · part b.ii · 3 marks

November 2023 Paper 2

Worked-example retry · Attempt part b.ii only.

(c) Explain the result for tube 8.

(3)

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(Total for Question 7 = 15 marks)

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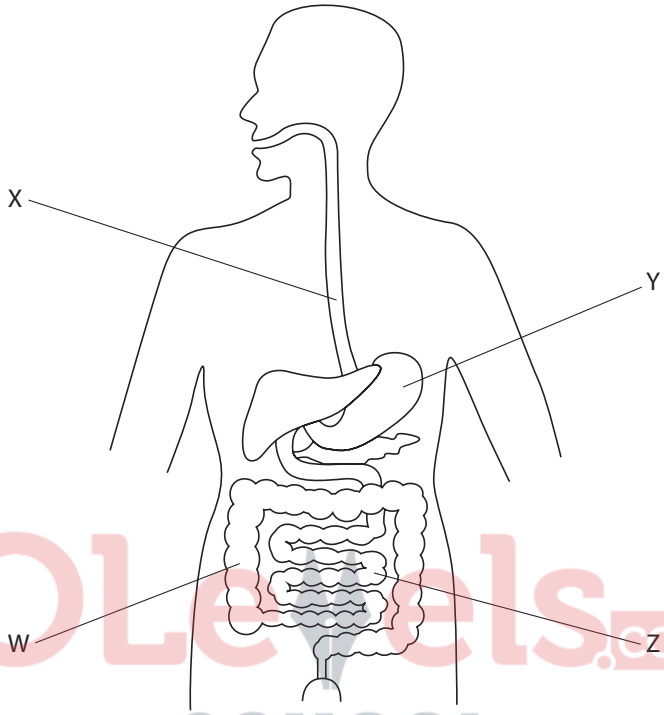
Working and answer

2. Question 2 · part b.iii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.iii only.

2 (a) The diagram shows the human digestive system.



(i) Name the structures labelled W, X, Y and Z.

(4)

W .....

X .....

Y .....

Z .....

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2. Question 2 · part b.iii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.iii only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

relax	coordination	contract	liquid
peristalsis	muscles	enzymes	bolus

Use words from the box to complete the passage.

(4)

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

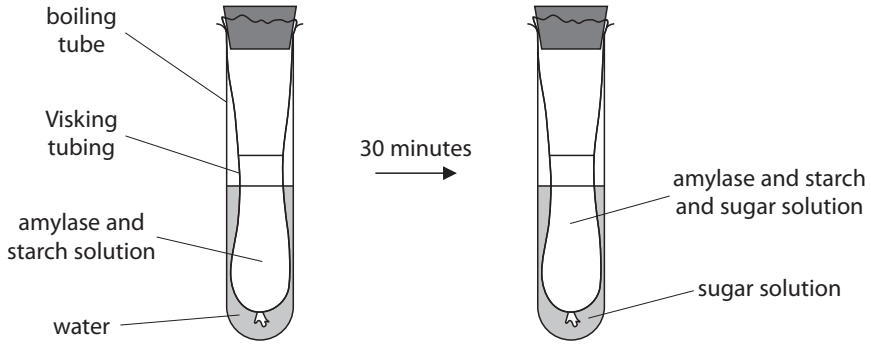
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2. Question 2 · part b.iii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b.iii only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.



(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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2. Question 2 · part b.iii · 2 marks · Working and answer

Working and answer

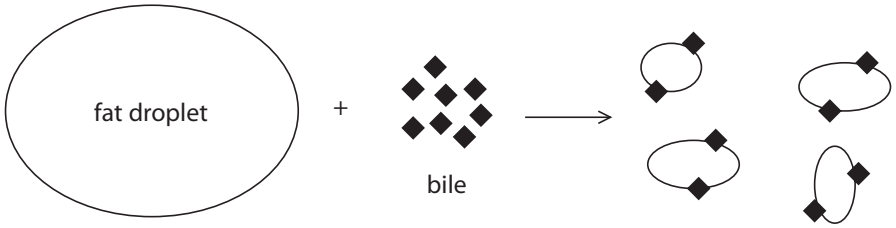
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3. Question 3 · part b.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.ii only.

3 (a) The diagram represents the effect of bile on a large fat droplet.



(i) Where is bile produced?

(1)

- ☒ **A** gall bladder
- ☒ **B** liver
- ☒ **C** small intestine
- ☒ **D** stomach

(ii) Describe the role of bile in the digestion of fat.

(3)

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3. Question 3 · part b.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.ii only.

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(b) Three food tests are used on a solution of food.

The table shows the reagent used and the colour of the food solution after each test.

Reagent used	Colour of the food solution after the test
iodine solution	blue/black
biuret solution	purple
Benedict's solution	blue

(i) Describe what the food tests show about the dietary components in the food.

(3)

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(ii) Saliva is added to a new sample of the food solution and the mixture is kept at 37 °C for 10 minutes.

The food tests are then repeated and the solution turns brick red when tested with Benedict's reagent.

Explain the effect of saliva on the food solution.

(3)

3. Question 3 · part b.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.ii only.

(c) Food is digested as it moves along the alimentary canal.

This process is helped by a diet containing fibre.

Explain how food is moved along the alimentary canal and how fibre helps this process.

(5)

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(Total for Question 3 = 15 marks)

3. Question 3 · part b.ii · 3 marks · Working and answer

Working and answer

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4. Question 2 · part b · 3 marks

November 2024 Paper 1

Further practice · Attempt part b only.

2 A student investigates the activity of an enzyme on fat found in milk.
This is the student's method.

- pour milk and enzyme solution into a test tube
- determine the pH of the solution every minute for 10 minutes

The table shows the student's results.

Time in minutes	pH of solution
1	6.8
2	6.4
3	6.0
4	5.8
5	5.6
6	5.2
7	5.0
8	4.8
9	4.6
10	4.6

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4. Question 2 · part b · 3 marks

November 2024 Paper 1

Further practice · Attempt part b only.

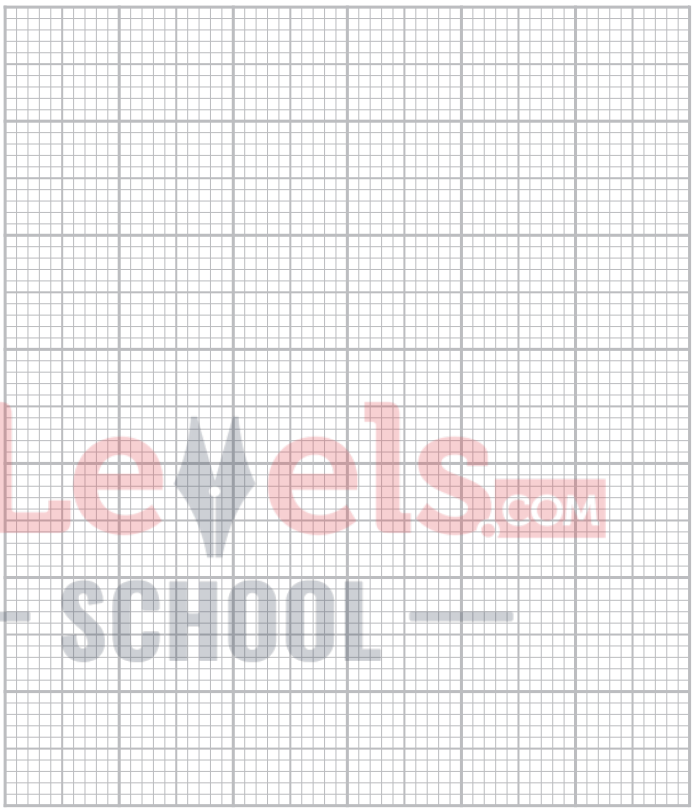
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(a) (i) Plot a graph of the student's results.

(3)



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(ii) Draw a line of best fit.

(1)

(iii) Give the trend shown by the graph.

(1)

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4. Question 2 · part b · 3 marks

November 2024 Paper 1

Further practice · Attempt part b only.

(b) Explain the difference in the pH at 1 minute and the pH at 10 minutes.

(3)

(c) Name the independent variable in this investigation.

(1)

(d) Give two variables that the student should control in this investigation.

(2)

1 .....

2 .....

(e) State how the student could improve the reliability of their results.

(1)

(Total for Question 2 = 12 marks)

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4. Question 2 · part b · 3 marks · Working and answer

Working and answer

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5. Question 4 · part b.i · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.i only.

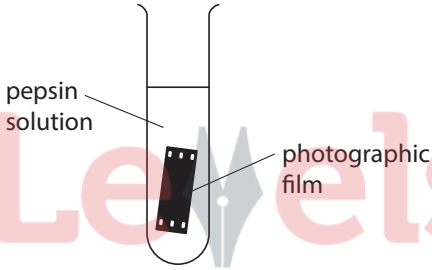
4 Photographic film is made of clear plastic, coated in a protein. The protein contains silver compounds which give the film its black colour.

Pepsin is an enzyme that digests protein.

A student investigates how the concentration of pepsin affects the time taken to digest the protein in a piece of photographic film.

This is the student's method.

- place a piece of photographic film into a test tube
- add a 1% pepsin solution
- measure the time for the film to become clear
- repeat the investigation using other concentrations of pepsin



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The table shows the student's results.

Percentage pepsin concentration (%)	Time taken for film to become clear in seconds
1	112
2	102
3	93
4	84
5	84
6	84

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5. Question 4 · part b.i · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.i only.

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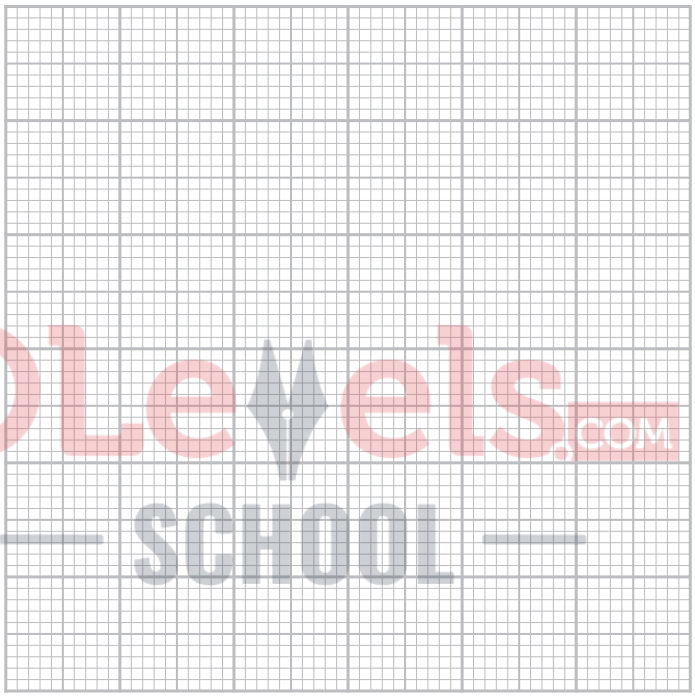
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(a) (i) Plot a graph of the results on the grid.

(4)

(ii) Draw a curve of best fit.

(1)



(b) (i) Explain why the photographic film becomes clear.

(3)

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5. Question 4 · part b.i · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b.i only.

(ii) Explain which would be the best pH to carry out this investigation.

(2)

(iii) State three variables that should be controlled during this investigation other than pH.

(3)

1 .....

2 .....

3 .....

(c) Suggest why the time taken for the film to become clear is the same for 4%, 5% and 6% concentrations of pepsin.

(2)

(Total for Question 4 = 15 marks)

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5. Question 4 · part b.i · 3 marks · Working and answer

Working and answer

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6. Question 7 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

7 Cheese is processed from milk. During this process, an enzyme is used to convert the lactose in milk into lactic acid.

A student uses this method to investigate the process.

- dissolve lactose in water to produce a solution
- put the lactose solution into three different test tubes
- keep the first test tube at 2 °C, the second test tube at 25 °C and the third test tube at 55 °C
- add enzyme solution to the lactose solution in each test tube
- measure the amount of lactic acid in the solution every 10 hours for a total of 60 hours

The table shows the student's results.

Time after adding enzyme in hours	Amount of lactic acid in the solution in arbitrary units		
	2 °C	25 °C	55 °C
0	0.1	0.1	0.1
10	0.1	0.2	0.2
20	0.1	0.3	0.2
30	0.2	0.8	0.4
40	0.3	1.8	0.7
50	0.4	2.2	1.0
60	0.6	2.3	1.3

(a) Describe what additional information another student would need to repeat this investigation.

(4)

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6. Question 7 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

(b) (i) Estimate the optimum temperature for this enzyme.

(1)

(ii) Explain why it is not possible to give an accurate value for the optimum temperature using the student's method.

(2)

(c) The student continued the investigation for a further 60 hours.

Explain the results that would be obtained.

(3)

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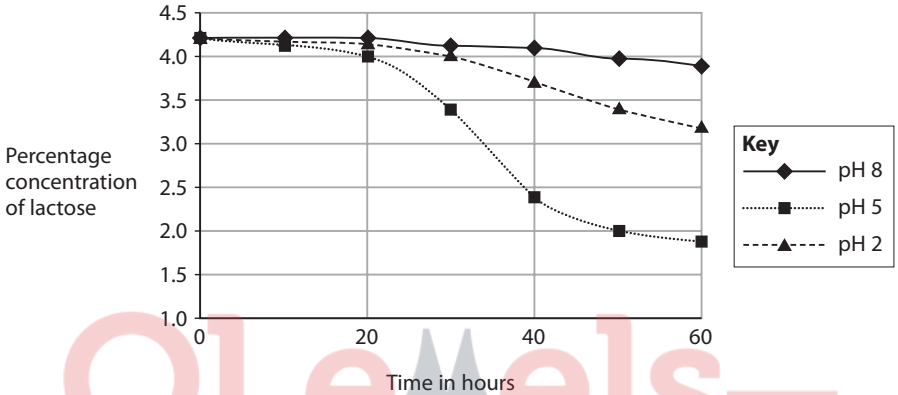
6. Question 7 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

(d) The student uses the same enzyme to investigate the effect of pH on the conversion of the lactose into lactic acid.

The student's results are shown in the graph.



The student wants to determine the most effective conditions for the conversion of lactose into lactic acid in cheese manufacture.

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6. Question 7 · part c · 3 marks · Working and answer

Working and answer

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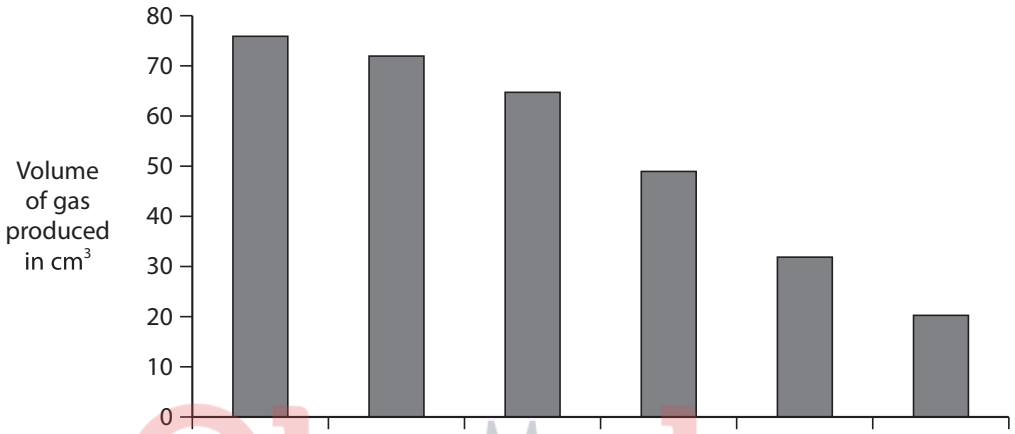
7. Question 3 · part a.iv · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iv only.

3 A student investigates a reaction. The reaction is catalysed by an enzyme and produces a gas. The student collects the gas for 10 minutes and measures the volume of the gas. The student repeats this every 10 minutes for one hour.

The bar chart shows the student's results.



(a) (i) Add a suitable labelled scale to the *x*-axis.

(2)

(ii) Estimate the total volume of gas produced after 30 minutes.

(2)

volume of gas = ... cm^3

(iii) Describe the trend shown by the results.

(3)

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P 7 2 5 8 5 A 0 6 2 0

7. Question 3 · part a.iv · 3 marks

June 2023 Paper 1

Further practice · Attempt part a.iv only.

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(iv) Explain the student's results obtained in the investigation.

(3)

(b) Explain the effect on the results if the student had added a competitive inhibitor at the start of the investigation.

(2)

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(Total for Question 3 = 12 marks)

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7. Question 3 · part a.iv · 3 marks · Working and answer

Working and answer

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8. Question 4 · part c · 2 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

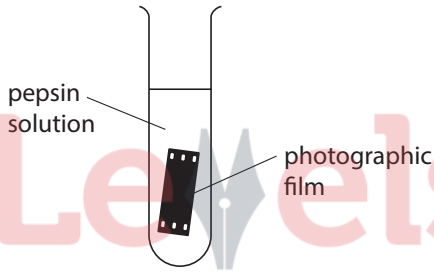
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8. Question 4 · part c · 2 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

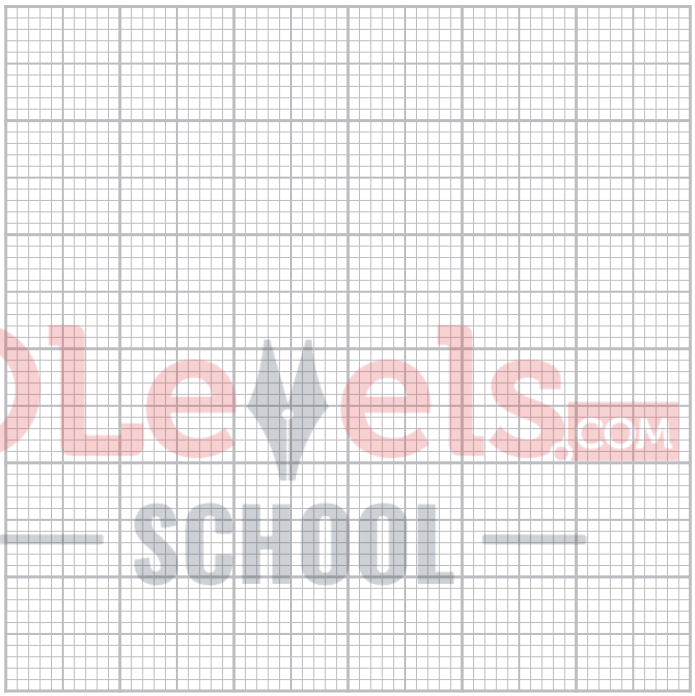
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(a) (i) Plot a graph of the results on the grid.

(4)

(ii) Draw a curve of best fit.

(1)



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(b) (i) Explain why the photographic film becomes clear.

(3)

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P 7 4 6 2 8 A 0 9 2 0

8. Question 4 · part c · 2 marks

June 2025 Paper 1R

Further practice · Attempt part c only.

(ii) Explain which would be the best pH to carry out this investigation.

(2)

(iii) State three variables that should be controlled during this investigation other than pH.

(3)

1 .....

2 .....

3 .....

(c) Suggest why the time taken for the film to become clear is the same for 4%, 5% and 6% concentrations of pepsin.

(2)

(Total for Question 4 = 15 marks)

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8. Question 4 · part c · 2 marks · Working and answer

Working and answer

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